

Design and Construction of a Microcontroller Based Railway Reservation Management System

Abstract

The Railway Reservation Management System is a computer-based application designed to automate the booking, management, and cancellation of train tickets, replacing the traditional manual process that is often slow, tedious, and prone to human error, especially as the volume of passengers continues to grow across the rail network. The system is built around a centralized database that stores detailed information on trains, routes, schedules, station lists, fares, and registered passengers, and it allows a passenger to search for available trains between specified source and destination stations, view real-time seat availability across different classes such as sleeper, AC, and general, reserve a seat by entering personal and journey details, generate a printable ticket carrying a unique Passenger Name Record (PNR) for tracking, and cancel an existing booking whenever necessary. Every reservation or cancellation instantly updates the seat inventory in the database, which prevents problems such as double booking or overselling of seats, while a waiting list mechanism automatically promotes and confirms pending tickets as confirmed seats are freed up through cancellations. Fares are computed automatically based on factors such as distance travelled and class of travel, removing the need for manual fare calculation and reducing the chance of billing errors. Beyond the passenger-facing functions, the system includes a dedicated administrative module that grants a system administrator the privilege to add or remove trains, update schedules, routes and fares, monitor and search booking records, and generate useful reports on passenger traffic, occupancy, and revenue over a given period, while ordinary users remain restricted to booking, viewing, and cancelling only their own tickets, ensuring proper data security and separation of roles. Altogether, the system was developed following a structured approach involving requirement analysis, system design, and implementation, using a relational database for reliable data storage and a simple, user-friendly interface for interaction, and the finished project offers a reliable, efficient, and easy-to-use platform that significantly reduces the time spent booking tickets, minimizes human error, eliminates seat duplication, and equips railway authorities with an effective tool for reservation management and reporting, with clear potential for further enhancement through features such as online payment integration, SMS/email notifications, and mobile application accessibility.

Submitted By:

Team Members:

AKSHAY PUTTA (25B11CS022)

J. DILEEP KUMAR (25B11CS342)

SAMPATHI HEMANTH (25B11CS844)

SASANK AKULA (25B11CS027)